

General info:

Date: 20/08/2025

Report period: 08/08/2025 – 22/08/2025

BIOFILM2 – Michelle Kuttel

Byron Clarke, Ruth Rudolph, Sonali Naidoo

Task activities during the report period

Week 3 submission: Prototype

- Ruth created the skeleton code of the classes for the Simulation, and Visualization using the class diagrams
- Ruth started work on the visualization
- Byron started work on the simulation: made the first bacteria and made it move
- Sonali will write code to add basic bacterial growth and EPS production to the program

Week 3 submission: Progress reports

- Byron created this document, and wrote the general topics of the report, and filled in the “Task activities during the report period”, “Planned activities for the next report period”, “problems”, and “Revised schedule and goals section” section
- Ruth will write the details for the visualization in the “Planned activities for the next report period” section
- Sonali and Ruth will review and possibly revise the problems section of this report

Planned activities for the next report period

The simulation

Simulation module:

- Making the bacteria grow and produce EPS
- Bacteria and EPS movement calculations
- Writing code to run these operations in parallel
- Intermediate output algorithm including cluster calculations

Visualization:

- Reading in the visualization file (output from the simulation)
- Displaying the component on which the simulation will be viewed

- Showing a bacterium on the surface
- General class details for visual bacteria properties

Graphing:

- Decide on whether to use Python or if there's an alternative Java framework
- Make a method for each graph type

The final report

- Review all requirements for the final report, find dependencies for when it can be written, and then delegate parts of it between the group, and write it.

Problems

The previous report's planned activities for this report period were successfully completed in this report period.

- Class diagram to actual code disparity

Many methods were left out of the class diagrams that later were found to be necessary in development. These will be cautiously added as necessary while maintaining the original intended functionality of the originally planned methods intact.

- Testing during development

Due to the nature of the project, there are not many use cases. In addition, the parallelism and concurrency may pose an issue when testing during the development stage, as certain aspects of the code may not be complete yet prompting additional work in the form of mocking.

Revised schedule and goals

No major changes warranting a revised schedule or new goals occurred.